Mohammad Nehad Moghrabi

AI & Machine Learning Engineer | LLM Applications | Backend Software Engineer

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Professional Summary

Computer Science and Artificial Intelligence graduate (MTI, June 2026) with a machine learning engineering internship at ICARDA and a public portfolio of AI, systems, and backend projects. Built a GPT-style transformer inference engine in x86-64 assembly, xray, a 10-language code-analysis tool published on npm, and ARGUS, a security assessment platform with evidence-gated LLM analysis. Builds tested services in Python, TypeScript, and Go with PostgreSQL, Redis, Docker, and CI. Based in Cairo, available immediately.

Technical Skills

AI & Machine Learning (ML): Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, NumPy, Pandas, deep learning, transfer learning, model evaluation, k-fold cross-validation, hyperparameter optimization, SHAP; geospatial ML with Google Earth Engine, GIS, remote sensing, and satellite imagery.

LLMs & NLP: Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), LLM agents, LangChain, Hugging Face Transformers/Diffusers, Qwen, FAISS, embeddings, BM25, reranking, semantic search, prompt engineering, guardrails, LoRA fine-tuning, Stable Diffusion, transformers.

Backend & Architecture: Object-Oriented Programming (OOP), system design, clean architecture, FastAPI, NestJS, Fastify, Gin, Node.js, REST APIs, OpenAPI, WebSockets, JWT/OTP authentication, RBAC, BullMQ, Celery.

Data & Storage: SQL, PostgreSQL/PostGIS, SQLite, Redis, Prisma, S3-compatible storage, ETL, database migrations.

Languages & Frontend: Python, TypeScript, JavaScript, Go, Rust, C, x86-64 Assembly (NASM), SQL, HTML/CSS; React, Next.js, Vite, MapLibre GL, VS Code extension API.

Cloud, DevOps & Testing: AWS (EC2, S3), Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Git, pnpm, Turborepo, Vitest, Jest, Playwright, Python unittest, integration and end-to-end testing.

Systems & Security: TCP/IP, HTTP/1.1, BSD sockets, multithreading, SIMD (AVX2/FMA), static analysis, tree-sitter, graph algorithms (PageRank, Tarjan SCC, Dijkstra, Louvain), penetration testing, SSRF defenses.

Professional Experience

International Center for Agricultural Research in the Dry Areas (ICARDA)
Machine Learning Engineer Intern *Aug 2025 – Sep 2025*

- Built an end-to-end rainfall prediction pipeline using Random Forest regression in Google Earth Engine, improving R-squared by approximately 15% over baseline regressors.
- Developed, trained, and evaluated XGBoost and TensorFlow deep neural network models in Python for geospatial prediction using k-fold cross-validation and hyperparameter optimization.
- Applied SHAP to quantify feature importance and produce model interpretability reports.
- Processed multispectral satellite imagery with NDVI, EVI, NDWI, and LST indices for vegetation, water, and land-temperature analysis.

Projects

xray: Codebase Architecture Mapper
TypeScript, Node.js, tree-sitter, SQLite, Fastify, React, Vite, Docker, GitHub Actions, Vitest, Playwright

- Published on npm a static-analysis tool that maps a repository into a graph of symbols, calls, imports, inheritance, and HTTP routes across 10 languages (TypeScript compiler API; tree-sitter for Python, Go, Java, Kotlin, Rust, C#, PHP, Ruby).
- Engineered graph analyses for impact and blast-radius risk scoring, import-cycle detection with suggested cuts (Tarjan SCC), call routes (Dijkstra and Yen k-shortest paths), BM25F search re-ranked by personalized PageRank, Louvain clustering, and architecture diffs between git revisions.
- Designed an incremental SQLite-backed index: cold scans of FastAPI (128K lines) in 3.5 s and Zod (104K lines) in 13.4 s, with unchanged rescans in 11–137 ms across 5 benchmark repositories.
- Shipped a CLI, React canvas UI, VS Code extension, Dockerized self-hosted service, two GitHub Actions (architecture checks, PR impact comments), and a gallery of 16 mapped repositories; 17-package monorepo with 200+ Vitest tests, precision/recall scoring on 13 golden fixtures, and Playwright e2e tests in CI.

ARGUS: Authorized AI Security Assessment Platform
TypeScript, NestJS, Next.js, BullMQ, Prisma, PostgreSQL, Redis, Qwen, Docker, Vitest
- Architected a multi-tenant platform for assessing explicitly authorized web targets: NestJS API, Next.js dashboard, and BullMQ worker orchestrating crawling, OpenAPI/GraphQL discovery, fingerprinting, and 9 deterministic scanners (TLS, CORS, headers, cookies, auth).
- Enforced scope with SSRF and DNS-rebinding defenses; added JWT, RBAC, Argon2, tenant isolation, integrity-hashed evidence, 100+ unit and security tests, and hardened Docker containers with CI.
- Integrated a private Qwen LLM with prompt boundaries and schema and grounding validation, falling back to deterministic findings when the model is unavailable; generated evidence-backed reports.

PotatoLM: Transformer Language Model in x86-64 Assembly
x86-64 NASM Assembly, AVX2/FMA SIMD, Python, NumPy, Make, Linux
- Wrote a GPT-style transformer inference engine in about 5,000 lines of NASM assembly: causal multi-head attention, LayerNorm, sampling, per-layer KV cache, BPE tokenizer, and a terminal chat.
- Optimized inference with AVX2/FMA kernels, runtime CPU feature detection, and a scalar fallback; on a 0.9M-parameter model, the KV cache cut 128-byte generation from 2.39 s to 0.043 s (56×) and AVX2 ran 4.4× faster than scalar.
- Implemented backpropagation and Adam in assembly, checked against a NumPy reference trainer; trained a BPE tokenizer from scratch (8,192 tokens); 61 tests cover assembly-vs-NumPy parity, finite-difference gradients, and SIMD parity.

GeoFlow: No-Code Geospatial Analysis Platform
Python, FastAPI, Next.js, PostgreSQL/PostGIS, Redis, Celery, Docker, MapLibre GL
- Created a Dockerized GIS platform with a Python raster engine (clipping, reprojection, statistics, Cloud Optimized GeoTIFF exports), validated node-based workflows, user-owned projects, and rotating-token authentication.

Home Food Delivery Marketplace Backend
Go, Gin, PostgreSQL, Redis, OpenAPI, Docker, S3-compatible storage
- Developed a contract-first backend with OTP/JWT authentication, rotating refresh tokens, RBAC, KYC onboarding, clean architecture, modular-monolith boundaries, reversible SQL migrations, and 180+ automated tests.

Cosmetics E-Commerce Platform
NestJS, PostgreSQL, Prisma, Redis, Docker, React, TanStack Start
- Delivered a bilingual English/Arabic platform with authentication, RBAC, checkout, inventory, audit logging, Redis-backed queues and caching, and WebSocket updates, gated by Jest, Vitest, Playwright, and accessibility checks.

HTTP/1.1 Servers from Raw TCP
C, Rust, TCP/IP, BSD sockets, multithreading
- Implemented HTTP/1.1 servers without web frameworks: request parsing, routing, keep-alive, chunked and streaming bodies, static files, a worker pool with a bounded queue, graceful shutdown, and real-socket integration tests.

Education

Modern University for Technology and Information (MTI) — **Cairo, Egypt**
BSc in Computer Science and Artificial Intelligence — *Oct 2022 – Jun 2026*
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Linear Algebra for Machine Learning.

Certifications

- IBM Data Science Professional Certificate, IBM / Coursera
- Data Scientist Professional Certificate, DataCamp
- Linear Algebra for Machine Learning and Data Science, DeepLearning.AI / Coursera
- GIS, Mapping, and Spatial Analysis Specialization, University of Toronto / Coursera
- Cisco Certified Network Associate (CCNA), Cisco
- CompTIA Network+, CompTIA

Languages

Arabic: Native | English: Fluent (Professional Proficiency)